# WASP-10b

**R-1807**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_140821` · created 2026-08-03T03:05:53+00:00

**PRACTICE / VALIDATION RUN — not submitted to any science database**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2456890.7993 +/- 0.0024 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.195 +/- 0.059 |
| Transit depth (Rp/Rs)^2 | 3.8 +/- 2.32 [%] |
| Orbital Inclination (inc) | 87.54 +/- 1.87 |
| Airmass coefficient 1 (a1) | 0.89 +/- 0.49 |
| Airmass coefficient 2 (a2) | 0.11 +/- 1.22 |
| Scatter in the residuals of the lightcurve fit is | 60.21 % |
| Best Comparison Star | #1 - [265, 329] |
| Optimal Aperture | 0.87 |
| Optimal Annulus | 3.75 |
| Transit Duration (day) | 0.083 +/- 0.021 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.195 ± 0.059 vs 0.1592 (deviation 0.61σ) |
| Duration fit vs published | 0.083 d vs 0.0946 d (deviation 0.55σ) |
| Mid-time O–C | -4.4 min |
| Depth SNR | 3.3 |
| Residual scatter | 60.21 % |

## Light curve

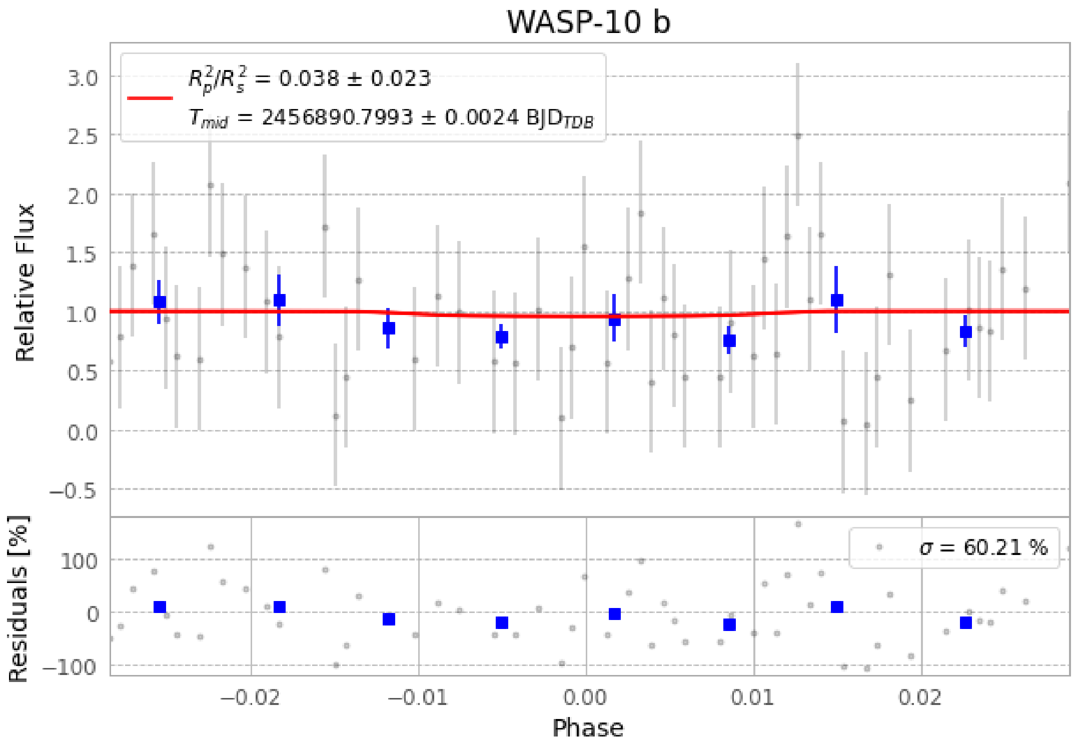


Figure 1 — FinalLightCurve_WASP-10 b_2014-08-20.png (SHA-256 52501a1f4eed1d51…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [564, 103], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c2826bc3b53e3ba2…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

## Data provenance

94 FITS frames (62 MB), `WASP-10140821045412.FITS` → `WASP-10140821093312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-140821.log (3fb63b430d36…), FinalLightCurve_WASP-10 b_2014-08-20.png (52501a1f4eed…), FinalLightCurve_WASP-10 b_2014-08-20.csv (57e90042738d…)

## Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)